ESP Lab1: Prep Sheet

Signal Modulation by Decimation and Interpolation

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Table 1: Configuration of Labgroup 1

Lab group	$M = L$	**Chirp [kHz]**	**Length**	f_{bat}	f_s **ADC, DAC**	x_λ **at CH0 / CH1**
ESPL/01	3	18 . . . 21	2.5 s	18 kHz	50 kHz, 50 kHz	$\lambda = 0$ / $\lambda = 2$

3 Preparation

Use of generative AI

Tool: Claude (Anthropic), October 2026. The lab instructions and the Moodle files were provided as context. Essential prompts (translated from German, condensed):

1. "Write a prompt that I can use to work on the preparation tasks of ESP Lab 1."
2. "Lab group ESPL/01: $M = L = 3$, chirp 18–21 kHz, 2.5 s, $f_{bat} = 18$ kHz, $f_s = 50$ kHz, polyphases $\lambda = 0$ / $\lambda = 2$. Create a commented MATLAB script for sections 3.1–3.5 with all required plots, using `firpmord`/`firpm` and `write_coeff` as in the tutorial script. Explain for each section why the result is correct, using concrete frequencies. Extend `dec_int_standalone.c` to $M = L = 3$ and explain why `g_PolyWUpdate` is necessary. Explain every non-trivial step so that I can explain the code in the lab."
3. "Create a LaTeX text file of the Word prep sheet template with listings for the MATLAB code."

3.1 Creating a Chirp Signal

Add your results, including **Matlab code** and **figures**, to the preparation sheet. **Explain** why you think the result is correct.

Listing 1: MATLAB code, parameters and Section 3.1

```
%% ESP Lab 1 - Preparation: Signal Modulation by Decimation and Interpolation
% Lab group ESPL/01
%   M = L = 3, chirp 18 ... 21 kHz, length 2.5 s, f_bat = 18 kHz
%   fs (ADC, DAC) = 50 kHz, polyphases at DAC: CH0 -> lambda = 0, CH1 -> lambda = 2
%
% Requirements: Signal Processing Toolbox (chirp, spectrogram, firpmord, firpm, freqz)
%               write_coeff.m (Moodle) in the same folder
%
% Run section by section (Ctrl+Enter) or the whole script (F5).

clear; close all; clc;

%% Parameters (Table 1, ESPL/01)
fs     = 50e3;      % sampling frequency ADC and DAC [Hz]
M      = 3;         % downsampling factor
L      = 3;         % upsampling factor (M = L)
f0     = 18e3;      % chirp start frequency [Hz]
f1     = 21e3;      % chirp stop frequency [Hz]
T      = 2.5;       % chirp length [s]
f_bat  = 18e3;      % bat sound frequency [Hz]
fs_low = fs/M;      % low sampling frequency f_s' = 16.667 kHz

% identical spectrogram settings for all plots -> figures are comparable
Nwin  = 1024;       % window length (Hamming)
Nover = 768;        % overlap (75 %)
Nfft  = 1024;       % FFT length

% section for the stem plots: n = n1 ... n2 (60 samples, far from the start)
n1 = 2000;
n2 = 2059;

%% 3.1 Chirp signal x[n]
n = 0:round(T*fs)-1;            % sample index n = 0 ... 124999
t = n/fs;                        % time axis [s]
x = chirp(t, f0, T, f1);         % linear chirp, f(t) = f0 + (f1-f0)*t/T, amplitude 1

figure('Name', '3.1 Chirp x[n]');
subplot(2,1,1);
spectrogram(x, hamming(Nwin), Nover, Nfft, fs, 'yaxis');
title(sprintf('3.1  Spectrogram X[l,k]: chirp %g ... %g kHz, f_s = %g kHz', ...
f0/1e3, f1/1e3, fs/1e3));

subplot(2,1,2);
sel = (n >= n1) & (n <= n2);     % logical index of the section
stem(n(sel), x(sel), 'filled', 'MarkerSize', 3); grid on;
xlabel('n'); ylabel('x[n]');
title(sprintf('3.1 Time Domain Signal x[n] for n = %d ... %d ', n1, n2));

```

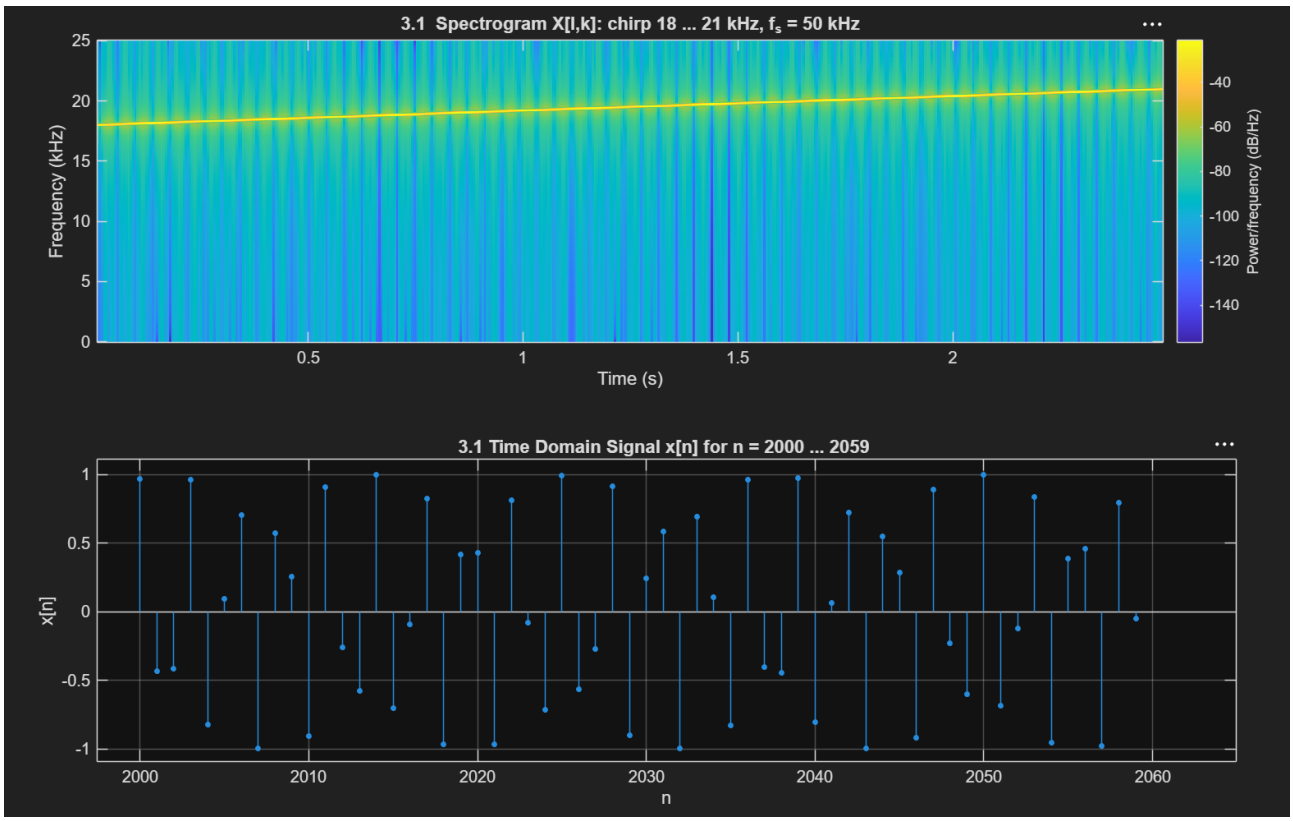


Figure 1: Chirp signal $x[n]$: spectrogram $X[l,k]$ (top) and stem plot of the section $n = 2000, \ldots, 2059$ (bottom).

Explanation: The chirp is generated on the time axis $t = n/f_s$ with `chirp(t, f0, T, f1)` (Listing 1). The spectrogram in Fig. 1 shows exactly this: a single straight line from 18 kHz to 21 kHz over 2.5 s. This limitation in time spreads a small part of the signal energy over all frequencies (spectral leakage), far below the chirp itself. At n = 2000 the chirp frequency is about 18 kHz. With $f_s =$ 50 kHz there are fewer than three samples per period ($50/18 \approx 2.78$), so the oscillation of the chirp cannot easily be recognised by the human eye. Instead, a low-frequency sine-like pattern is visible: every third sample lies on a slowly varying curve, because three samples correspond to slightly more than one full period of the chirp.

3.2 Downsampling and Polyphase Representation

Add your results, including **Matlab code** and **figures**, to the preparation sheet. **Explain** why you think the result is correct.

Listing 2: MATLAB code, Section 3.2

```
%% 3.2 Downsampling by M and polyphase signals x_lambda[m] = x[m*M + lambda]
Nshow = 100;                    % first 100 samples of each polyphase
m     = 0:Nshow-1;              % low-rate index
xpoly = cell(1, M);             % xpoly{lambda+1} = x_lambda[m]

figure('Name', '3.2 Polyphase signals');
for lambda = 0:M-1
% MATLAB indices start at 1: index 1+lambda corresponds to n = lambda,
% step M takes every M-th sample -> n = lambda, lambda+M, lambda+2M, ...
xpoly{lambda+1} = x(1+lambda : M : end);

subplot(M, 1, lambda+1);
stem(m, xpoly{lambda+1}(1:Nshow), 'filled', 'MarkerSize', 3); grid on;
ylabel(sprintf('x_{%d}[m]', lambda));
title(sprintf('3.2  Polyphase \\lambda = %d:  x_{%d}[m] = x[%dm + %d],  f_s'' = %.3f kHz', ...
```

```
        lambda, lambda, M, lambda, fs_low/1e3));
        end


        % expected aliased frequency in the low-rate domain (fold into 0 ... fs'/2)
        falias = @(f) abs(f - round(f/fs_low)*fs_low);
        fprintf('3.2: chirp %g ... %g kHz appears at %.3f ... %.3f kHz after downsampling\n', ...
        f0/1e3, f1/1e3, falias(f0)/1e3, falias(f1)/1e3);

```

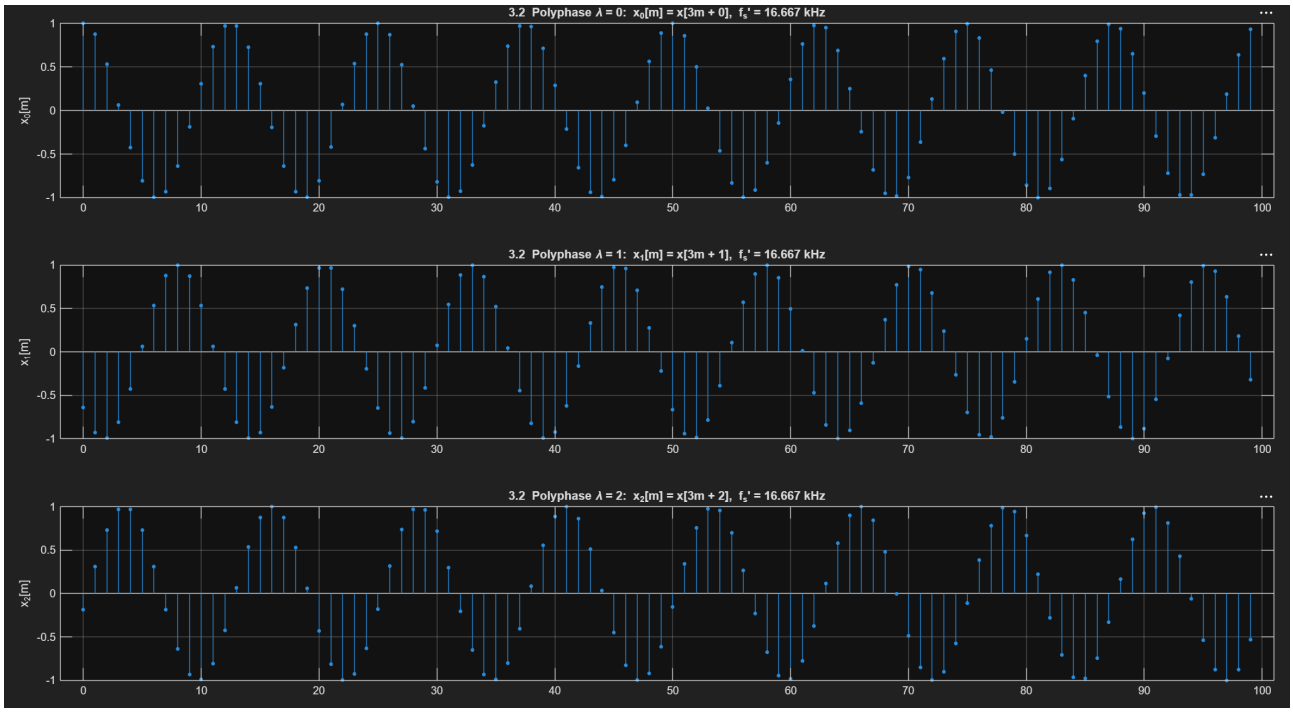


Figure 2: Polyphase signals $x_\lambda[m] = x[3m + \lambda]$, $\lambda = 0, 1, 2$, first 100 samples at the low sampling frequency $f_s' = f_s/3 = 16.67\,\text{kHz}$.

Explanation. The three subplots in Fig. 2 show the three downsampled signals (polyphases) $x_\lambda[m]$. Each polyphase takes every third sample of $x[n]$, starting at $n = \lambda$.

All three polyphases oscillate with the same low frequency of about 1.33 kHz (about 12.5 samples per period). Since no anti-aliasing filter is applied, the chirp at 18 kHz lies above $f_s'/2$ and is aliased to

$$f' = f - f_s' = 18\,\text{kHz} - 16.67\,\text{kHz} \approx 1.33\,\text{kHz}.$$

This is the slow pattern already visible in the stem plot of Section 3.1.

3.3 Upsampling without Interpolation Filter

Add your results, including **Matlab code** and **figures**, to the preparation sheet. Explain the **shape of the spectrogram** and provide reasons **based on specific frequencies**. Assume that a **bat sound of** f_{bat} (as given in Table 1 for your ESPL group) is fed into the system's ADC. Is it **reproduced at the DAC output**? If so, at **what frequency** / frequencies?

Listing 3: MATLAB code, Section 3.3

```
        %% 3.3 Upsampling of x_0[m] by L WITHOUT interpolation filter
        x0 = xpoly{1};                      % polyphase lambda = 0
        yu = zeros(1, L*length(x0));        % y'[n] at high rate fs
        yu(1:L:end) = x0;                   % y'[nL] = x_0[n], L-1 zeros in between

```

```
        figure('Name', '3.3 Upsampler output y''[n]');
        subplot(2,1,1);
        spectrogram(yu, hamming(Nwin), Nover, Nfft, fs, 'yaxis');
        title(sprintf('3.3  Spectrogram Y''[l,k]: x_0[m] upsampled by L = %d, no interpolation filter'
    , L));

        subplot(2,1,2);
        nu  = 0:length(yu)-1;
        sel = (nu >= n1) & (nu <= n2);
        stem(nu(sel), yu(sel), 'filled', 'MarkerSize', 3); grid on;
        xlabel('n'); ylabel('y''[n]');
        title(sprintf('3.3  y''[n] for n = %d ... %d  (every %d-th sample non-zero)', n1, n2, L));

        % where does a bat tone f_bat appear at the DAC output?
        fa    = falias(f_bat);                          % frequency in the low-rate domain
        f_img = [fa + (0:L-1)*fs_low, (1:L)*fs_low - fa];   % all images k*fs' +/- fa
        f_img = unique(f_img(f_img <= fs/2));           % only 0 ... fs/2 is visible
        fprintf('3.3: f_bat = %g kHz -> %.3f kHz after downsampling\n', f_bat/1e3, fa/1e3);
        fprintf('     upsampler output (0 ... %g kHz): %s kHz, each with amplitude 1/L = %.1f dB\n',
    ...
        fs/2e3, mat2str(round(f_img)/1e3), 20*log10(1/L));

```

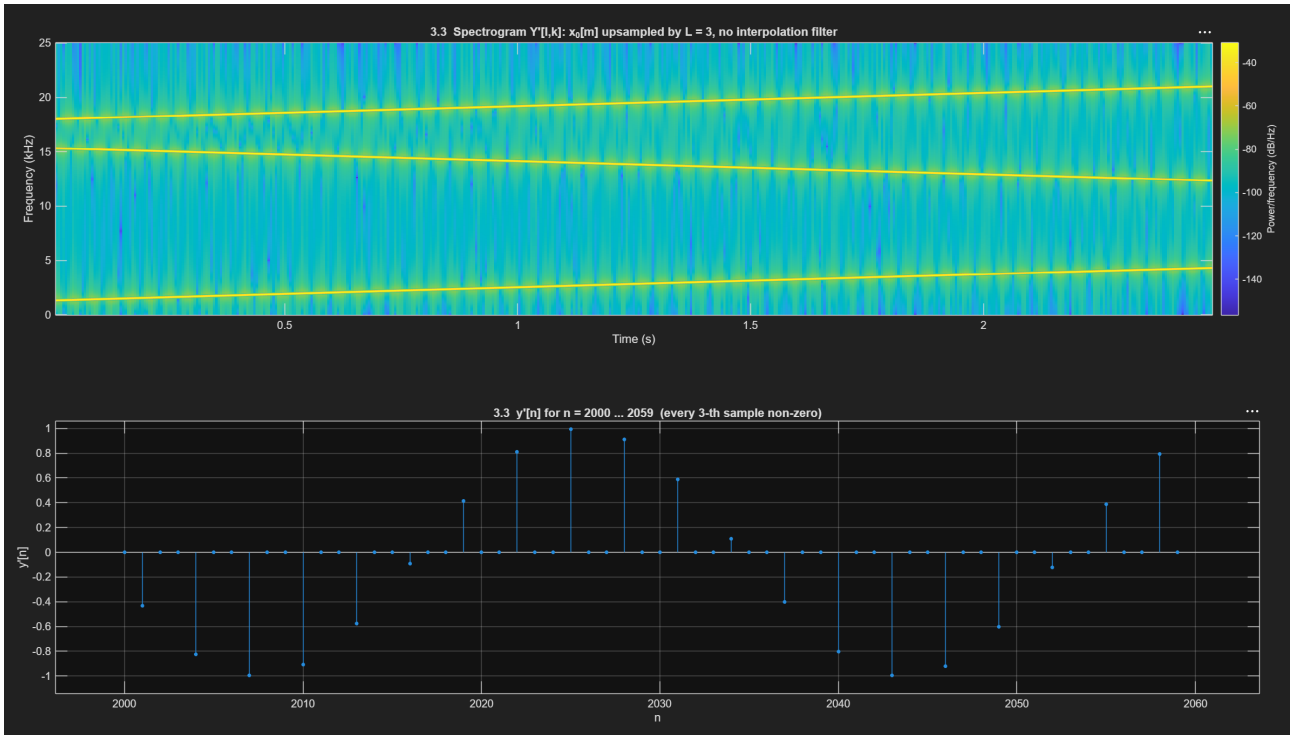


Figure 3: Upsampler output $y'[n]$ ($L = 3$, no interpolation filter): spectrogram $Y'[l, k]$ (top) and stem plot for $n = 2000, \ldots, 2059$ (bottom).

Explanation. Upsampling inserts $L - 1 = 2$ zeros between the samples of $x_0[m]$, which brings the signal back to the high sampling rate f_s (see stem plot, every third sample is non-zero). The zeros do not restore the original signal: the spectrum of $x_0[m]$ is periodic with $f_s' = 16.67\,\text{kHz}$, and all periods within $0 \ldots f_s/2$ become visible as images. The spectrogram therefore shows three chirps:

- baseband f': $1.33\,\text{kHz} \to 4.33\,\text{kHz}$ (rising), the chirp aliased by downsampling,
- first image $f_s' - f'$: $15.33\,\text{kHz} \to 12.33\,\text{kHz}$ (falling, mirrored),
- second image $f_s' + f'$: $18\,\text{kHz} \to 21\,\text{kHz}$ (rising), at the position of the original chirp.

All three components match the non-zero samples equally well, so they have the same level, each reduced to $1/L$ ($-9.5\,\mathrm{dB}$).

A bat sound of $f_{bat} = 18\,\mathrm{kHz}$ is first aliased to $18 - 16.67 \approx 1.33\,\mathrm{kHz}$ by downsampling. At the DAC output it is reproduced at 1.33 kHz, 15.33 kHz and 18 kHz. It therefore appears at its original frequency of 18 kHz, but attenuated and accompanied by two additional tones.

3.4 Applying an Interpolation Filter

Add your results, including **Matlab code** and **figures**, to the preparation sheet. **Compare** the results from 3.3 and 3.4 and explain them.

Listing 4: MATLAB code, Section 3.4

```
%% 3.4 Interpolation (anti-imaging) filter
d_pass = 0.001;                 % pass-band ripple
d_stop = 0.001;                 % stop-band ripple (-60 dB)

% After upsampling the useful band of x_0 is 0 ... fs'/2 = fs/(2L).
% The first image starts at fs' - fs'/2 = fs/(2L) -> the stop band must start there.
f_stop = fs/(2*L);              % 8.333 kHz
f_pass = f_stop - 1e3;          % 7.333 kHz, transition width 1 kHz

[N_FIR, fo, mo, w] = firpmord([f_pass f_stop], [1 0], [d_pass d_stop], fs);
N_firpmord = N_FIR;             % order estimated by firpmord

% firpmord only ESTIMATES the order. The filter on the DSP uses 16-bit
% coefficients, so check the quantised filter and increase N if necessary.
while true
b = L * firpm(N_FIR, fo, mo, w);     % gain L compensates the 1/L of zero insertion
b = round(b*32768)/32768;            % Q15 quantisation exactly as on the DSP
[H, fH] = freqz(b, 1, 2^15, fs);
ripple  = max(abs(abs(H(fH <= f_pass))/L - 1));        % relative to gain L
att_dB  = -20*log10(max(abs(H(fH >= f_stop)))/L);      % relative to gain L
if ripple <= d_pass && att_dB >= -20*log10(d_stop)
break;
end
N_FIR = N_FIR + 1;
end
fprintf('3.4: f_pass = %.3f kHz, f_stop = %.3f kHz\n', f_pass/1e3, f_stop/1e3);
fprintf('     firpmord order %d, final order %d (%d coefficients)\n', ...
N_firpmord, N_FIR, length(b));
fprintf('     quantised filter: pass-band ripple %.5f, stop-band attenuation %.1f dB\n', ...
ripple, att_dB);

figure('Name', '3.4 Interpolation filter');
plot(fH/1e3, 20*log10(abs(H)/L)); grid on; hold on;
xline(f_pass/1e3, '--', 'f_{pass}'); xline(f_stop/1e3, '--', 'f_{stop}');
yline(-60, ':', '-60 dB');
xlim([0 fs/2e3]);               % 0 ... 25 kHz
xticks(0:1:fs/2e3);             % tick every 1 kHz
xlabel('f [kHz]'); ylabel('|H(f)| / L [dB]');
title(sprintf('3.4  Interpolation filter, order %d, Q15 coefficients, gain L = %d', N_FIR, L))
;
ylim([-100 5]);

% apply the filter to the upsampler output
y = filter(b, 1, yu);

figure('Name', '3.4 Interpolation filter output y[n]');
subplot(2,1,1);
spectrogram(y, hamming(Nwin), Nover, Nfft, fs, 'yaxis');
title('3.4  Spectrogram Y[l,k]: after interpolation filter');

subplot(2,1,2);
```

```
        % longer section than in 3.1/3.3: the 1.333 kHz output has 37.5 samples per
        % period, 120 samples show about 3 periods
        n2_34 = n1 + 119;
        sel34 = (nu >= n1) & (nu <= n2_34);
        stem(nu(sel34), y(sel34), 'filled', 'MarkerSize', 3); grid on;
        xlabel('n'); ylabel('y[n]');
        title(sprintf('3.4  y[n] for n = %d ... %d  (group delay %.1f samples)', n1, n2_34, N_FIR/2));

```

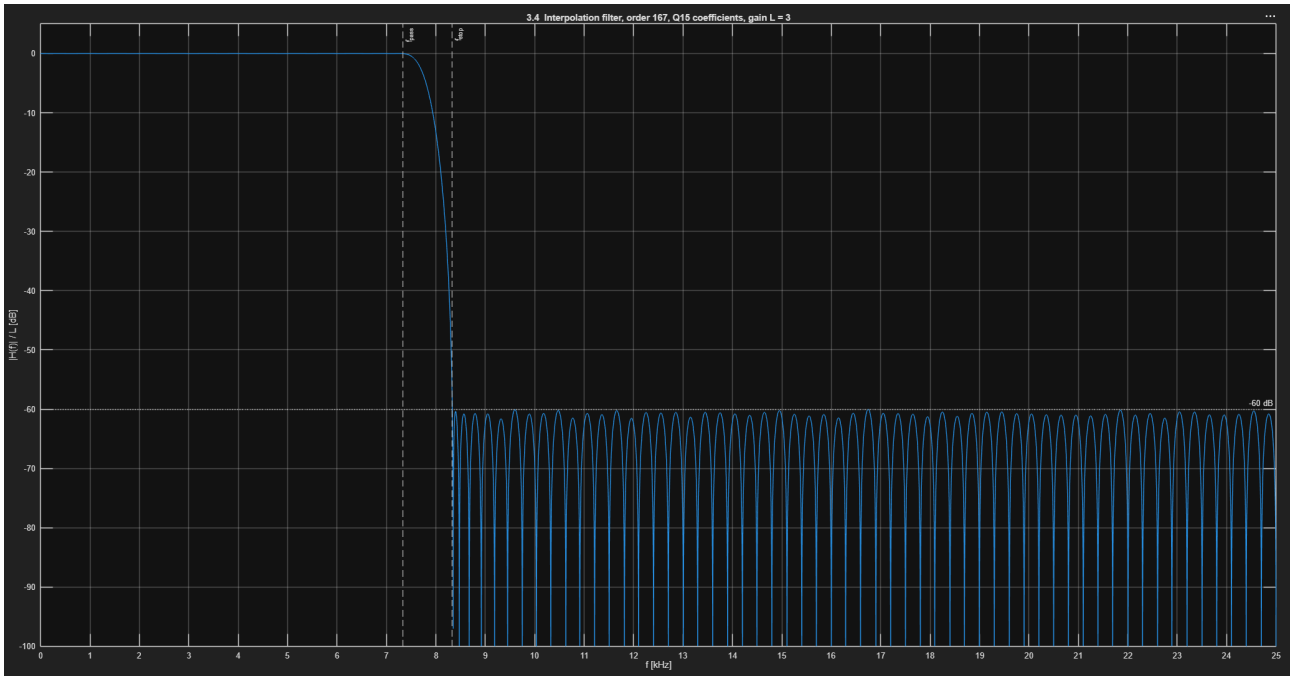


Figure 4: FIR-Filter normalized frequency response

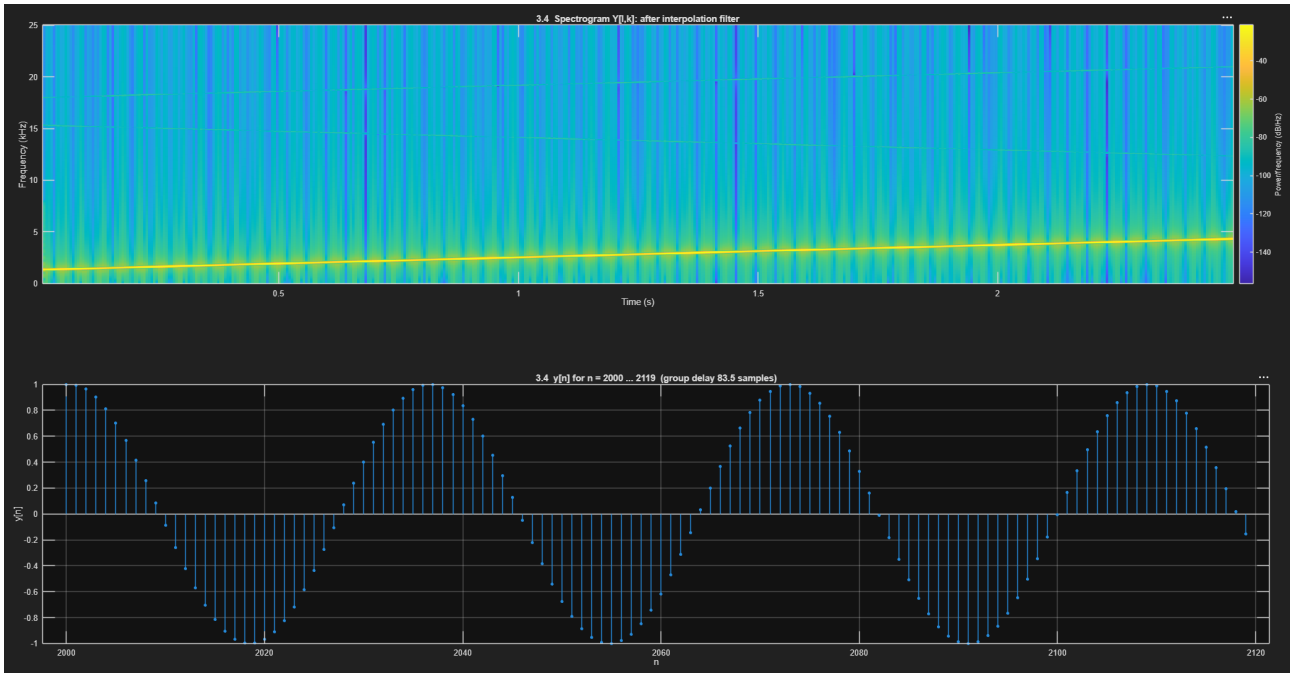


Figure 5: Interpolation filter output $y[n]$: spectrogram $Y[l,k]$ (top) and stem plot for $n = 2000, \ldots, 2119$ (bottom).

Explanation. The interpolation filter is a low-pass filter with $f_{pass} = 7.33\,\text{kHz}$ and $f_{stop} = f_s/(2L) =$

8.33 kHz. It keeps the baseband chirp (1.33 kHz → 4.33 kHz) and suppresses both images (15.33 kHz → 12.33 kHz and 18 kHz → 21 kHz). The images are still faintly visible because a real FIR filter of finite order does not attenuate them completely: the design specification $\delta_{stop} = 0.001$ corresponds to 60 dB, which matches the distance between the chirp and the remaining images in the spectrogram.

Compared to 3.3, the zeros in the time signal have been filled. In the frequency domain, only the baseband component remains, and a single sinusoid is non-zero at every sample. The stem plot therefore shows a smooth sine wave of about 1.33 kHz. Its amplitude is 1 again, because the filter has the gain $L = 3$ that compensates the factor $1/L$ caused by zero insertion.

3.5 Export Filter Coefficients to C

Generate the header file for the filter coefficients of your interpolation filter and bring the file with you to the lab session. **How many coefficients** does your FIR filter have? Which is the **largest and which the smallest** value? Do these values **fit into 16-bit format**? **Explain**.

Listing 5: MATLAB code, Section 3.5

```
%% 3.5 Export filter coefficients to a C header file
coeff_q15 = round(b*32768);                % integer values written by write_coeff
if any(coeff_q15 > 32767) || any(coeff_q15 < -32768)
error('Coefficient outside 16-bit range -> reduce gain or rescale!');
end

filename = 'interp_filter_L3.h';
file_ID  = fopen(filename, 'w');
fprintf(file_ID, '//---------------------------------------- \n');
fprintf(file_ID, '// designed with -- ESP_Lab1_Prep_ESPL01.m -- \n');
fprintf(file_ID, ['// ', date, '\n']);
fprintf(file_ID, '// interpolation filter, L = %d, gain = L\n', L);
fprintf(file_ID, '// Fs = %8.2f\n', fs);
fprintf(file_ID, '// fpass = %8.2f\n', f_pass);
fprintf(file_ID, '// fstop = %8.2f\n', f_stop);
fprintf(file_ID, '// delta_pass = %6.4f\n', d_pass);
fprintf(file_ID, '// delta_stop = %6.4f (%.0f dB)\n', d_stop, 20*log10(d_stop));
fprintf(file_ID, '// N_FIR = %d (firpmord estimate %d)\n', N_FIR, N_firpmord);
fprintf(file_ID, '//---------------------------------------- \n');
fprintf(file_ID, '#define N_DELAYS_INTERP %d\n', length(b));
fprintf(file_ID, 'short H_interp_delays[N_DELAYS_INTERP]; \n');
write_coeff(file_ID, 'b_interp', b, length(b));
fclose(file_ID);

fprintf('3.5: header %s written\n', filename);
fprintf('     number of coefficients: %d\n', length(coeff_q15));
fprintf('     largest  value: %6d  (%.4f)\n', max(coeff_q15), max(coeff_q15)/32768);
fprintf('     smallest value: %6d  (%.4f)\n', min(coeff_q15), min(coeff_q15)/32768);
fprintf('     sum of coefficients (DC gain): %.4f\n', sum(b));

```

Listing 6: generated C-Header with write_coeff.m

```
//----------------------------------------
// designed with -- ESP_Lab1_Prep_ESPL01.m --
// 31-Oct-2026
// interpolation filter, L = 3, gain = L
// Fs = 50000.00
// fpass =  7333.33
// fstop =  8333.33
// delta_pass = 0.0010
// delta_stop = 0.0010 (-60 dB)
// N_FIR = 167 (firpmord estimate 163)
//----------------------------------------
#define N_DELAYS_INTERP 168
short H_interp_delays[N_DELAYS_INTERP];
```

```
        short b_interp[168]={
            28,    -36,    -37,    -24,      5,     30,
            28,     -4,    -39,    -43,     -5,     45,
            60,     19,    -48,    -80,    -40,     45,
            100,     66,    -36,   -119,   -100,     19,
            136,    139,     10,   -146,   -183,    -50,
            148,    229,    103,   -137,   -276,   -170,
            110,    318,    249,    -63,   -352,   -340,
            -6,    372,    440,    100,   -372,   -544,
            -222,    345,    649,    374,   -285,   -748,
            -556,    183,    833,    770,    -30,   -896,
            -1016,   -187,    924,   1298,    483,   -904,
            -1621,   -887,    813,   2001,   1448,   -615,
            -2473,  -2275,    233,   3134,   3653,    548,
            -4303,  -6636,  -2665,   7860,  20754,  29572,
            29572,  20754,   7860,  -2665,  -6636,  -4303,
            548,   3653,   3134,    233,  -2275,  -2473,
            -615,   1448,   2001,    813,   -887,  -1621,
            -904,    483,   1298,    924,   -187,  -1016,
            -896,    -30,    770,    833,    183,   -556,
            -748,   -285,    374,    649,    345,   -222,
            -544,   -372,    100,    440,    372,     -6,
            -340,   -352,    -63,    249,    318,    110,
            -170,   -276,   -137,    103,    229,    148,
            -50,   -183,   -146,     10,    139,    136,
            19,   -100,   -119,    -36,     66,    100,
            45,    -40,    -80,    -48,     19,     60,
            45,     -5,    -43,    -39,     -4,     28,
            30,      5,    -24,    -37,    -36,     28,
        };

```

Listing 7: MATLAB command window output, Section 3.5

```
    3.5: header interp_filter_L3.h written
        number of coefficients: 168
        largest  value:  29572  (0.9025)
        smallest value:  -6636  (-0.2025)

```

Explanation. The FIR filter has 168 coefficients (order 167). The largest coefficient is 29572 (0.9025), the smallest is -6636 (-0.2025). The coefficients are symmetric, as expected for a linear-phase FIR filter.`write_coeff` converts the floating-point coefficients into the 16-bit Q15 format: each coefficient is multiplied by $2^{15} = 32768$ and rounded to an integer,

A 16-bit signed integer covers the range $-32768 \ldots 32767$, which corresponds to $-1 \leq h < 1$. Both the largest value 0.9025 and the smallest value -0.2025 lie within this range, so all coefficients fit into the 16-bit format, even though the filter contains the gain $L = 3$. On the DSP, the products of samples and coefficients are summed in a 32-bit accumulator and shifted back by 15 bits (`shft = 15`) to obtain a 16-bit output sample.

3.6 C Program with Interrupt Handlers

Extend the code for M and L values given in Table 1 for your ESPL group. The **polyphases** $x_{\lambda m}$ specified in Table 1 should be **fed to the respective DAC outputs**. **Why** is `g_PolyWUpdate` necessary? **Add your C code** to the preparation sheet.

Listing 8: Extended `dec_int_standalone.c` for $M = L = 3$ (CH0: $\lambda = 0$, CH1: $\lambda = 2$), Section 3.6

```c

#include <stdio.h>

#define M 3    // downsampling factor
#define L 3    // upsampling factor
#define LEN 30
#define ADC0 0
#define ADC1 1

// Global variable declaration
short int g_PolyW[M]={0,0}; // downsampled polyphase values
short int g_PolyWUpdate=0;   // last polyphase updated
short int g_i16AdcData[2]={0,0};  // 2 x 16 bit data of ADC input channels
short int g_i16DacData[2]={0,0};; // 2 x 16 bit data of DAC output  channels

short int g_SamplesX[30] = {
    1,2,3,4,5,6,7,8,9,10,
    11,12,13,14,15,16,17,18,19,20,
    21,22,23,24,25,26,27,28,29,30
}; // input values
short int g_SampleXIdx = 0; // index input samples

// decimation
void adcInt(void) {
    static short lambda = 0;
    // read values from ADC
    g_i16AdcData[ADC0] = g_SamplesX[g_SampleXIdx]; // next ADC sample
    g_i16AdcData[ADC1] = 0;                        // 2nd channel not used

    // create polyphase signals, low sampling rate
    switch (lambda) {
        case 0:
        g_PolyW[0] = g_i16AdcData[ADC0];
        g_PolyWUpdate = 0;
        break;
        case 1:
        g_PolyW[1] = g_i16AdcData[ADC0];
        g_PolyWUpdate = 1;
        break;
        case 2:
        g_PolyW[2] = g_i16AdcData[ADC0];
        g_PolyWUpdate = 2;
        break;
        default:
        break;
    }

    lambda = (lambda + 1) % M;
}

void dacInt(void) {
    // write values to DAC, high sampling rate
    switch (g_PolyWUpdate) {
        case 0:
        g_i16DacData[0] = g_PolyW[0];
        g_i16DacData[1] = 0;
        break;
        case 1:
        g_i16DacData[0] = 0; // both DAC output should be set to zero during polyphase = 1
        g_i16DacData[1] = 0;
        break;
        case 2:
        g_i16DacData[0] = 0;
        g_i16DacData[1] = g_PolyW[2];
        break;
```

```
        default:
        break;
    }
    // add interpolation filtering here
}

int main(void) {
    short int g_SamplesY[2][LEN]; // output values

    for (g_SampleXIdx = 0; g_SampleXIdx < LEN; g_SampleXIdx++) {
        dacInt(); // "call" of interrupt handler
        adcInt(); // "call" of interrupt handler

        // output values after ADC + DAC interrupt
        printf("g_PolyW[0]:%2d  g_PolyW[2]:%2d\n", g_PolyW[0], g_PolyW[2]);
        g_SamplesY[0][g_SampleXIdx] = g_i16DacData[0];
        g_SamplesY[1][g_SampleXIdx] = g_i16DacData[1];
    }

    // assemble polyphase signals y_0[n], y_1[1]
    printf("g_SamplesY[0]: ");
    for (g_SampleXIdx = 0; g_SampleXIdx < LEN; g_SampleXIdx++) {
        printf("%d ", g_SamplesY[0][g_SampleXIdx]);
    }
    printf("\n");

    printf("g_SamplesY[1]: ");
    for (g_SampleXIdx = 0; g_SampleXIdx < LEN; g_SampleXIdx++) {
        printf("%d ", g_SamplesY[1][g_SampleXIdx]);
    }
    printf("\n");

    return 0;
}

```

Listing 9: Simulated output of `adcInt()`/`dacInt()` for the input values 1 to 30 and both interrupt orders, Section 3.6

```
g_PolyW[0]: 1  g_PolyW[2]: 0
g_PolyW[0]: 1  g_PolyW[2]: 0
g_PolyW[0]: 1  g_PolyW[2]: 3
g_PolyW[0]: 4  g_PolyW[2]: 3
g_PolyW[0]: 4  g_PolyW[2]: 3
g_PolyW[0]: 4  g_PolyW[2]: 6
g_PolyW[0]: 7  g_PolyW[2]: 6
g_PolyW[0]: 7  g_PolyW[2]: 6
g_PolyW[0]: 7  g_PolyW[2]: 9
g_PolyW[0]:10  g_PolyW[2]: 9
g_PolyW[0]:10  g_PolyW[2]: 9
g_PolyW[0]:10  g_PolyW[2]:12
g_PolyW[0]:13  g_PolyW[2]:12
g_PolyW[0]:13  g_PolyW[2]:12
g_PolyW[0]:13  g_PolyW[2]:15
g_PolyW[0]:16  g_PolyW[2]:15
g_PolyW[0]:16  g_PolyW[2]:15
g_PolyW[0]:16  g_PolyW[2]:18
g_PolyW[0]:19  g_PolyW[2]:18
g_PolyW[0]:19  g_PolyW[2]:18
g_PolyW[0]:19  g_PolyW[2]:21
g_PolyW[0]:22  g_PolyW[2]:21
g_PolyW[0]:22  g_PolyW[2]:21
g_PolyW[0]:22  g_PolyW[2]:24
g_PolyW[0]:25  g_PolyW[2]:24
g_PolyW[0]:25  g_PolyW[2]:24
g_PolyW[0]:25  g_PolyW[2]:27
g_PolyW[0]:28  g_PolyW[2]:27
g_PolyW[0]:28  g_PolyW[2]:27
g_PolyW[0]:28  g_PolyW[2]:30
g_SamplesY[0]: 0 1 0 0 4 0 0 7 0 0 10 0 0 13 0 0 16 0 0 19 0 0 22 0 0 25 0 0 28 0
g_SamplesY[1]: 0 0 0 3 0 0 6 0 0 9 0 0 12 0 0 15 0 0 18 0 0 21 0 0 24 0 0 27 0 0
```

Explanation. The file `dec_int_standalone.c` for $M = L = 2$ was extended to $M = L = 3$ (Listing 8):

- `M` and `L` are set to 3, and `g_PolyW` has three elements.

- `adcInt()` gets the additional `case 2`. The input samples are distributed cyclically to $\lambda = 0, 1, 2$, i.e. $x_\lambda[m] = x[3m + \lambda]$ (decimation by $M = 3$).
- In `dacInt()`, `case 0` outputs $x_0[m]$ on CH0 and `case 2` outputs $x_2[m]$ on CH1. In all other DAC interrupts the channel receives a zero, so each polyphase value is followed by two zeros (upsampling by $L = 3$).

The output in Listing 9 confirms this: CH0 carries $1, 4, 7, \cdots = x[0], x[3], x[6], \cdots = x_0[m]$ and CH1 carries $3, 6, 9, \cdots = x[2], x[5], x[8], \cdots = x_2[m]$, each followed by two zeros. All values are delayed by one sample, because `main()` calls `dacInt()` before `adcInt()`, so the DAC outputs the value of the previous ADC interrupt. In the lab (Section 4.2), the bodies of `adcInt()` and `dacInt()` are inserted into the interrupt handlers of the `adda_simple` project, with `g_i16AdcData` and `g_i16DacData` replaced by the ADC and DAC registers.

Why is `g_PolyWUpdate` necessary? `g_PolyWUpdate` is the link between the two interrupt handlers: `adcInt()` writes which polyphase has just been updated, and `dacInt()` reads it to decide whether the current output sample is a polyphase value or an inserted zero. ADC and DAC interrupts are triggered independently, and their order cannot be predicted. A separate counter in `dacInt()` could therefore run out of phase with the ADC and output an old value or a value of the wrong polyphase. With `g_PolyWUpdate`, the interpolation stays synchronised with the decimation

The preparation is complete. Congratulations! **Please submit the preparation sheet in Moodle before the lab session.**